FINANCIAL PERFORMANCE ANALYSIS

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1. What problem was this project trying to solve?

The project aimed to analyze financial data to understand a company's sales performance, profitability, and business trends. Many organizations have large volumes of financial data but struggle to extract meaningful insights from it.

This project helps solve that problem by using data analytics techniques to identify:

- Which products generate the most profit
- Which countries contribute the most sales
- How discounts impact profitability
- Sales trends over time

The goal was to transform raw financial data into actionable business insights for better decision-making.

2. What dataset was used to solve the problem?

The dataset used in this project is the Microsoft Financial Sample Dataset.
This dataset contains financial transaction records including the following fields:

- Segment
- Country
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price
- Gross Sales
- Discounts
- Sales
- Cost of Goods Sold (COGS)
- Profit
- Date

The dataset is widely used for business intelligence and financial analysis projects.

3. How was the raw data prepared for analysis?

Before performing analysis, the raw dataset was cleaned and prepared using data preprocessing techniques. The following steps were performed:

- Removed blank rows and unnecessary records
- Checked for duplicate entries
- Standardized column names
- Converted date fields into proper date format
- Ensured numeric columns such as Sales, Profit, and Units Sold were correctly formatted
- Verified data consistency across all columns

These steps ensured the dataset was accurate, clean, and ready for analysis.

4. How was the data model designed?

A data model was created in Power BI to organize the data for analysis.
A star schema model was implemented with:
Fact Table

- Financial Data (Sales, Profit, Units Sold, Discounts)

Dimension Tables

- Product Table
- Country Table
- Segment Table
- Date Table

Relationships were created between the fact table and dimension tables using one-to-many relationships, allowing accurate filtering and aggregation across the dashboard.

5. How were key business metrics calculated?

Several key business metrics (KPIs) were calculated using SQL queries and Power BI measures. Important metrics include:

Total Sales
Total Sales = SUM(Sales)

Total Profit
Total Profit = SUM(Profit)

Total Units Sold
Total Units Sold = SUM(Units Sold)

Profit Margin
Profit Margin = Total Profit / Total Sales

For example, a calculated profit margin of 0.14 represents 14% profitability.

These metrics help measure the company's financial performance and efficiency.

6. What solution was developed to present the insights?

A Power BI interactive dashboard was developed to present the analysis results in a clear and visual format.

The dashboard includes:

- KPI Cards (Total Sales, Profit, Profit Margin, Units Sold)
- Sales by Country visualization
- Profit by Product chart
- Sales by Segment pie chart
- Monthly Sales Trend line chart
- Discount impact analysis
- Interactive filters (Product, Country, Segment)

This dashboard allows users to quickly explore financial performance and business trends.

7. What insights were discovered from the analysis?

The analysis revealed several important insights:

- Some products generate significantly higher profits than others
- Certain countries contribute a larger share of total revenue
- Higher discount levels reduce overall profitability
- Sales vary across different business segments
- Monthly sales trends show fluctuations across the year

These insights help businesses optimize pricing strategies, improve product performance, and focus on profitable markets.

8. What tools and technologies were used?

The following tools and technologies were used in the project:

- SQL – Data querying and analysis
- Power BI – Data visualization and dashboard creation
- Python (Pandas) – Data cleaning and preprocessing
- Excel / CSV Dataset – Data storage format

These tools are commonly used in real-world data analytics and business intelligence projects.

9. What was the final outcome of the project?

The final outcome of the project was a fully functional financial performance analysis dashboard built using Power BI.

The project successfully:

- Cleaned and prepared raw financial data
- Built a structured data model
- Calculated important financial metrics
- Developed an interactive dashboard for visualization

This project demonstrates how data analytics tools can transform raw financial data into meaningful insights that support business decision-making.

Thank You

for your time
and attention